

Sina Sajadmanesh

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SUMMARY

Senior AI Engineer specializing in multimodal foundation models, with a strong focus on translating research into high-performance, production-ready systems. Proven track record in building scalable model training pipelines and optimizing inference across heterogeneous hardware.

EDUCATION

Swiss Federal Institute of Technology (EPFL) , PhD in Electrical Engineering	Aug 2023
Sharif University of Technology , MSc in Information Technology Engineering	Sept 2016
University of Isfahan , BSc in Computer Software Engineering	Feb 2014

EXPERIENCE

Sony AI , Senior AI Engineer – Zurich, Switzerland	Jan 2026 – present
- Optimized and deployed ML models on Qualcomm edge devices using QAIRT / SNPE / QNN, extending official support to modern LLMs and VLMs - Architected and maintained a modular ML acceleration and deployment framework supporting diverse hardware targets (Nvidia GPUs, Qualcomm NPUs, Apple Silicon) - Developed compact vision-language models supporting image/video understanding and localization tasks, achieving competitive performance with reduced computational requirements	
Sony AI , AI Engineer – Zurich, Switzerland	Oct 2023 – Dec 2025
- Designed a modular and flexible geometric optimization algorithm, enabling NeurIPS'25 spotlight paper - Led end-to-end optimization of Sony's Argus VFM inference stack, resulting in up to $61\times$ speedup across 17 vision tasks - Developed 7 computer vision models, including object detection, depth estimation, super-resolution, and OCR, achieving SOTA or competitive results - Architected and implemented a foundational MTL framework for Sony's Argus VFM, supporting 17 vision tasks, enabling CVPR'25 publication	
Idiap Research Institute , Research Assistant – Martigny, Switzerland	May 2019 – Aug 2023
- Worked on differentially private machine learning with graph neural networks 3 papers in top-tier conferences (CCS, USENIX Security, and WSDM) with 340+ citations 7 invited talks at top universities and research institutions, including Imperial College, UIC, and Twitter 6 open-source projects with 130+ stars on GitHub 1 short course taught on "Trustworthy Machine Learning" at Artificial Intelligence Doctoral Academy - Finalist in CSAW Applied Research Competition for the best paper award in computer security in Europe	
Brave Software , Research Intern – Remote	Mar 2022 – May 2022
Developed a novel privacy-preserving federated learning framework for neural bandit models under client heterogeneity using PyTorch, Flower, Dask, and Tensorflow-Lite	
Sharif University of Technology , Research Assistant – Tehran, Iran	Nov 2014 – May 2019
- Worked on privacy-preserving machine learning, web data science, and information network analysis 4 papers in top-tier venues (WWW, TKDD, IoTJ, ASONAM) with 490+ citations	

- 5+ press coverages in prominent media outlets, including MIT Technology Review, France 24, and The Independent
- 2 open-source projects with 20+ stars on GitHub
- 1 semester course taught on "Fundamentals of Programming with Python"

SKILLS AND EXPERTISE

Foundation & Generative Models: Vision & Multimodal Models (ViT, CLIP, LLaVA) · Large Language Models (LLaMA, GPT, Qwen) · Parameter-Efficient Fine-Tuning (LoRA) · Self-Supervised Learning (MAE, DINO)

Model Optimization & Deployment: TensorRT · ONNX · Qualcomm QNN / SNPE / QAIIR · Apple CoreML / MLX · Optimum · Quantization · Graph Optimization · Performance Profiling

Machine Learning Frameworks: PyTorch · Hugging Face (Transformers, PEFT, Accelerate, Datasets) · OpenMMLab (MMDetection, MMEval, MMDetection)

Engineering & Tooling: Python · C++ · Bash · SQL · Data Processing (NumPy, Pandas, torchvision) · Environment & Packaging (uv, ruff) · CI/CD (Docker, GitHub Actions)

COMMUNITY AND PROFESSIONAL SERVICE

Invited Speaker: Imperial College London (2023, 2020), University of Illinois at Chicago (2022), L3S Research Center (2022), GNN User Group Meetup (2021), Twitter ML Seminar (2021)

Organizing / Program Committee: Privacy and Fairness in AI for Health (2023), PPAI (2024), WiseML (2023), PAIR2Struct (2022), DPML (2021)

Reviewer: NeurIPS (2026, 2024), CVPR (2026), TIFS (2025), IMWUT (2024), TDSC (2023), LoG (2023, 2022), AISTATS (2023), AIJ (2022), TBD (2021), TIST (2020), SNAM (2020), WWWJ (2018)

SELECTED PUBLICATIONS

Stella: Subspace Learning in Low-rank Adaptation using Stiefel Manifold Zhizhong Li, Sina Sajadmanesh, Jintao Li, Lingjuan Lyu NeurIPS 2025 🌟 Spotlight (top 3%)	Dec 2025
Argus: A Compact and Versatile Foundation Model for Vision Weiming Zhuang, Chen Chen, Zhizhong Li, Sina Sajadmanesh, et al. CVPR 2025	June 2025
ProGAP: Progressive Graph Neural Networks with Differential Privacy Guarantees Sina Sajadmanesh, Daniel Gatica-Perez WSDM 2024	Mar 2024
GAP: Differentially Private Graph Neural Networks with Aggregation Perturbation Sina Sajadmanesh, Ali Shahin Shamsabadi, Aurélien Bellet, Daniel Gatica-Perez USENIX Security 2023	Aug 2023
Locally Private Graph Neural Networks Sina Sajadmanesh, Daniel Gatica-Perez CCS 2021	Nov 2021
A Hybrid Deep Learning Architecture for Privacy-Preserving Mobile Analytics Seyed Ali Osia, Ali Shahin Shamsabadi, Sina Sajadmanesh, et al. IEEE Internet of Things Journal	May 2020